

GTU CSE 241 / 501 Quiz

Std Code:

Std Number:

2	1	0	1	0	4	0	0	4	0	1	7			
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class Digit{

public:

Digit();

Digit(unsigned int);

int check_digit(int) const;

friend ostream& operator<<(ostream& , const Digit&);

friend Digit& operator++(Digit&); // prefix

friend Digit operator++(Digit& , int); // postfix

friend ostream& operator<<(ostream& os ,const Digit& d);

friend const Digit operator+(const Digit& d1 , const Digit& d2);

private :

unsigned int digit;

};

```

Digit :: Digit : digit(0) {}

Digit :: Digit(unsigned int value) : Digit(check_digit(value)) {}

int Digit :: check_digit(int value) {
    if(value < 0 ){
        exit(1);
    }
    return (value % 10) ; }

ostream& operator<<(ostream& of , const Digit& d){
    of << d.digit <<;
    return of;
}

Digit& operator++(Digit& d) {
    d.digit +=1;
    d.digit = check_digit(d.digit);
    return d;
}

Digit operator++(Digit& d , int ignore){
    int temp =d.digit;
    d.digit++;
    d.digit = check_digit(d.digit);
    return Digit(temp); }

ostream& operator<<(ostream& os , const Digit& d){
    os << d.digit<<;
    return os; }

const Digit operator+(const Digit&d1 , const Digit& d2 ) {
    int temp = check_digit(d1.digit + d2.digit);
    return Digit(temp); }

```